

Align the target's graph views to a frozen source backbone with a learnable per-view contrastive loss — 2–4× faster MARL transfer, and continual learning scenario.

1. Why Transfer is hard in MARL

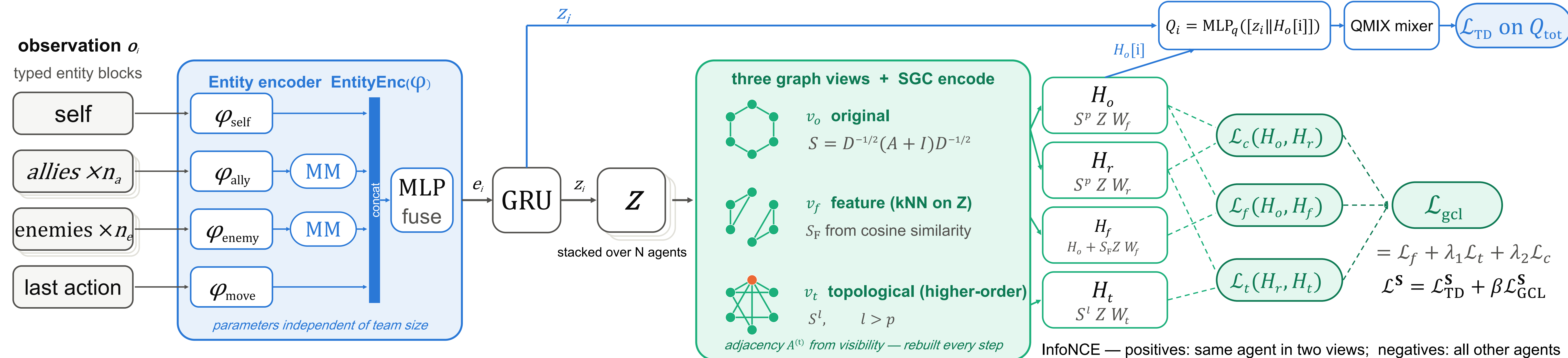
- Training cooperative agents from scratch for every new task is prohibitively sample-hungry.
- Source and target differ in team size, observation dimension and action spaces — naive weight transfer is architecturally ill-posed under population mismatch.
- Prior remedies retrain adapters per task (MALT), hand-design curricula (EPC), or never align representations (LA-QT) — inviting negative transfer.

2. Contributions

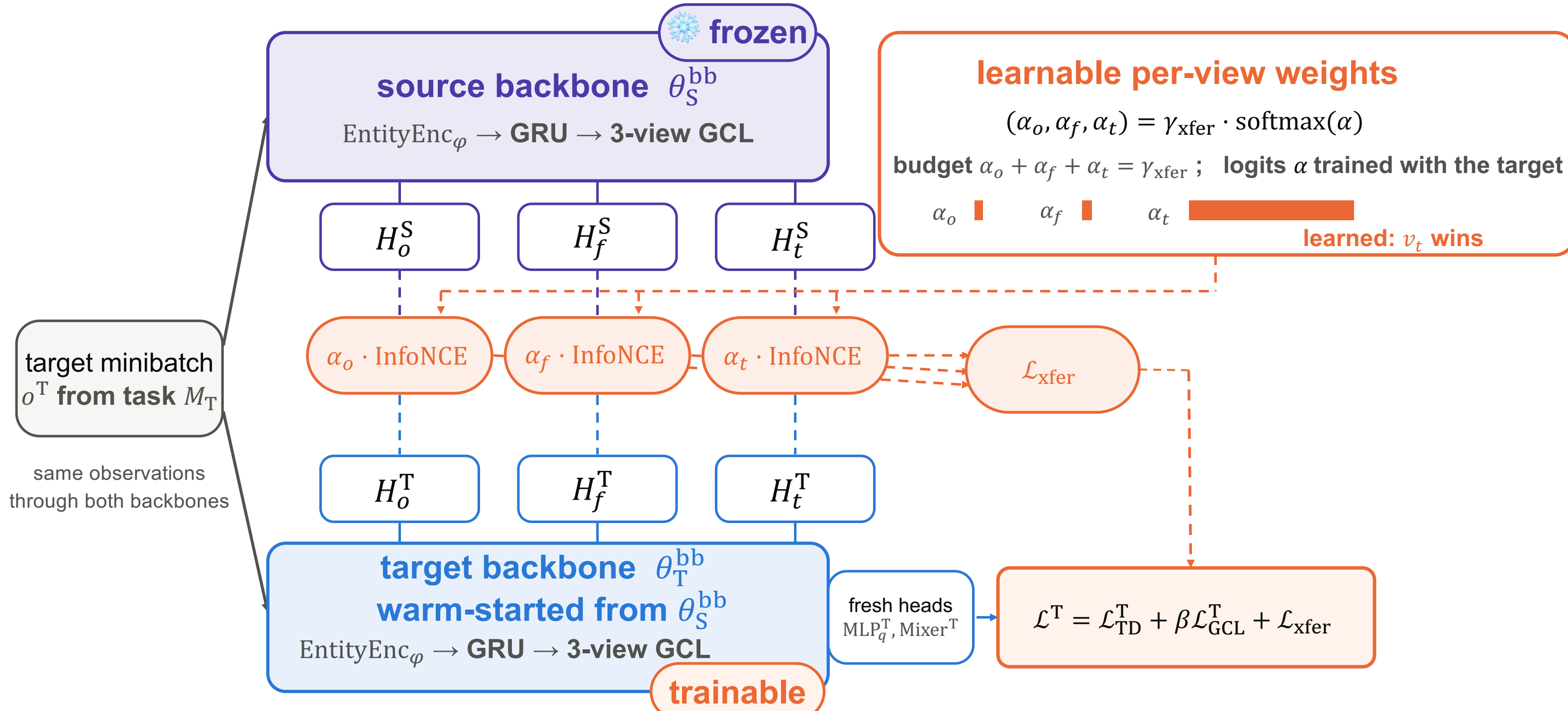
- ① Graph-contrastive transfer framework for cooperative MARL — re-purposes MAIL's multi-view GCL backbone (SGC on QMIX) as a transferable prior.
- ② Per-view, adaptively-weighted alignment loss — the model learns which contrastive view carries the transferable signal, per task pair.
- ③ A two-phase training protocol that chains into continual learning.

3. Method — Population-invariant Backbone, Per-view Alignment, Continual Chain

(a) Population-invariant multi-view graph-contrastive backbone source training $\mathcal{L}_S = \mathcal{L}_{TD} + \beta \mathcal{L}_{GCL}$

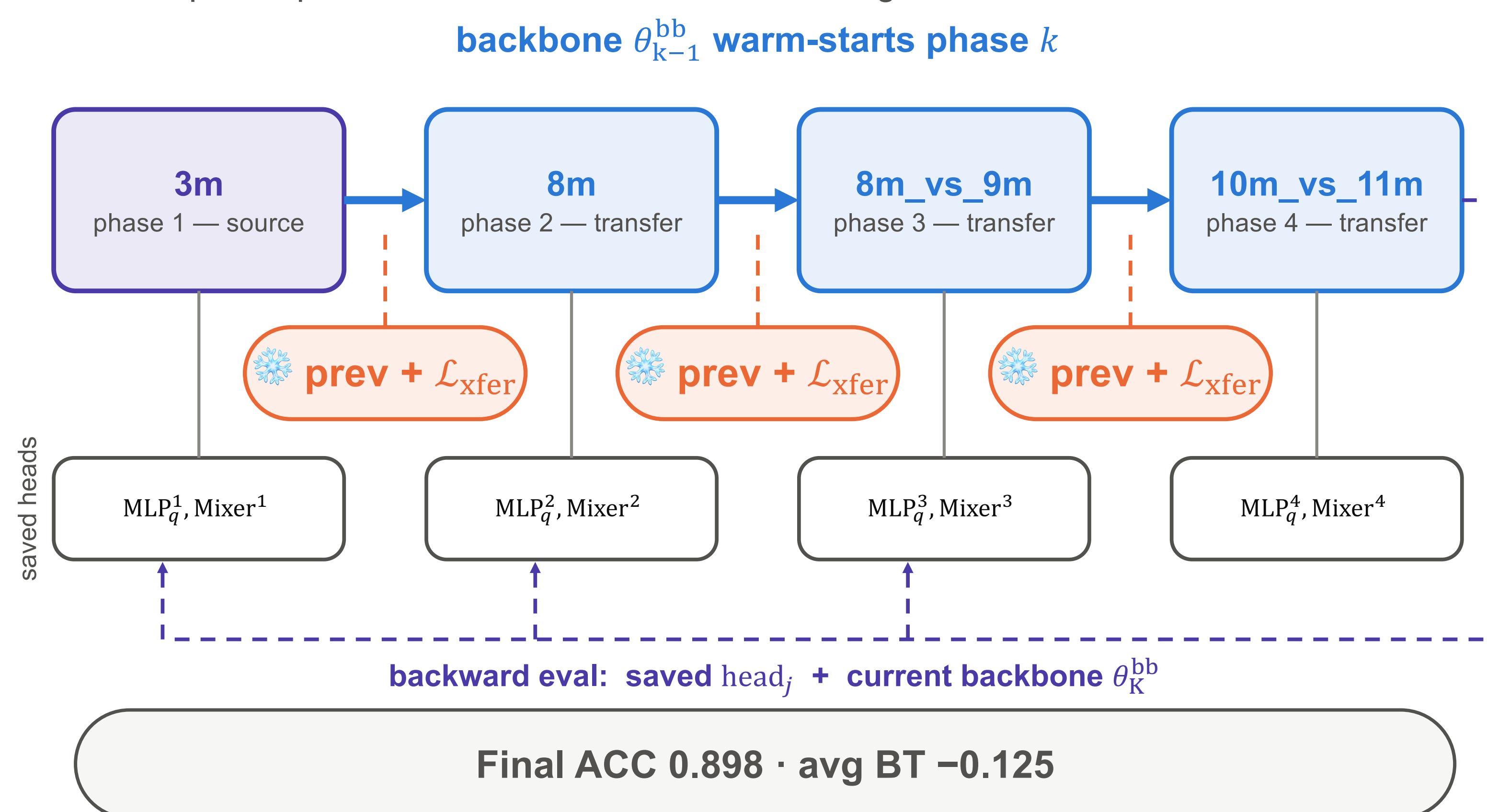


(b) Transfer (phase 2): per-view alignment to a frozen source — learnable α

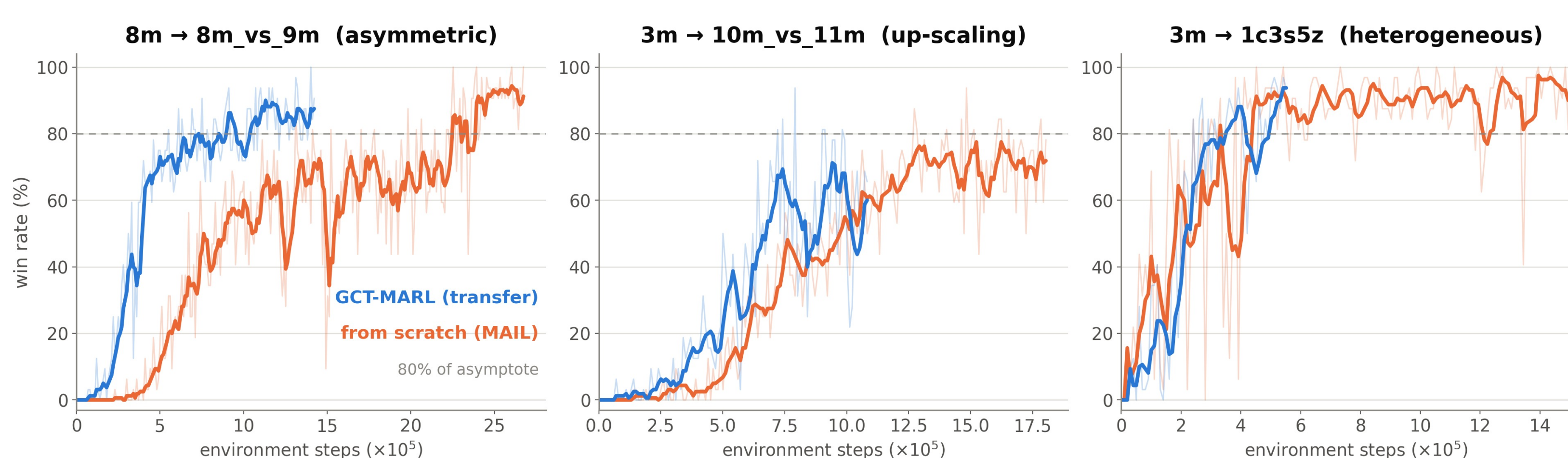


$$\mathcal{L}_{xfer} = \sum_{v \in \{o, f, t\}} \alpha_v \cdot \text{InfoNCE}(H_v^T, H_v^S), \quad \text{and } (\alpha_o, \alpha_f, \alpha_t) = \gamma_{xfer} \cdot \text{softmax}(\alpha)$$

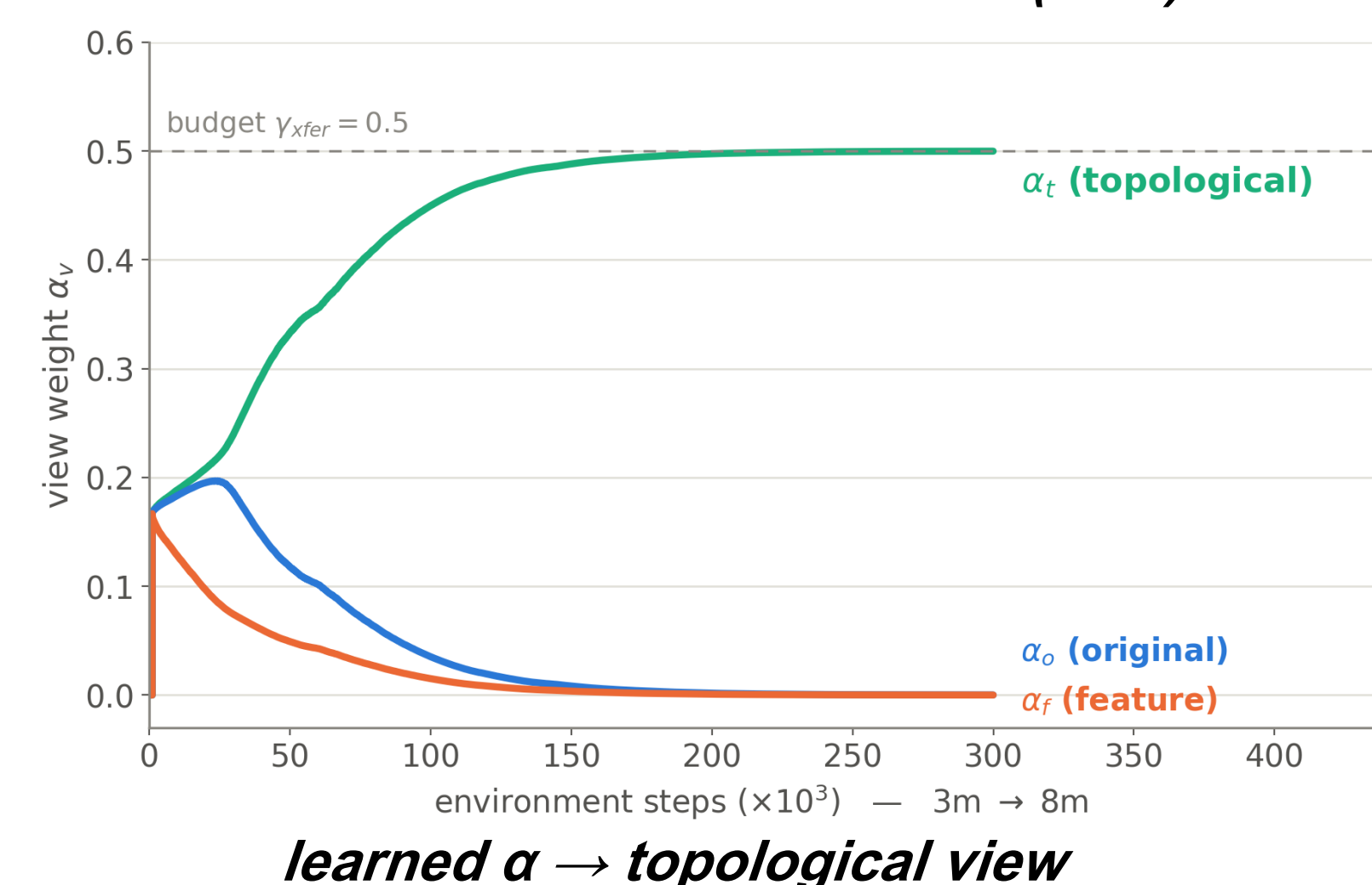
(c) Two-phase protocol chains into continual learning



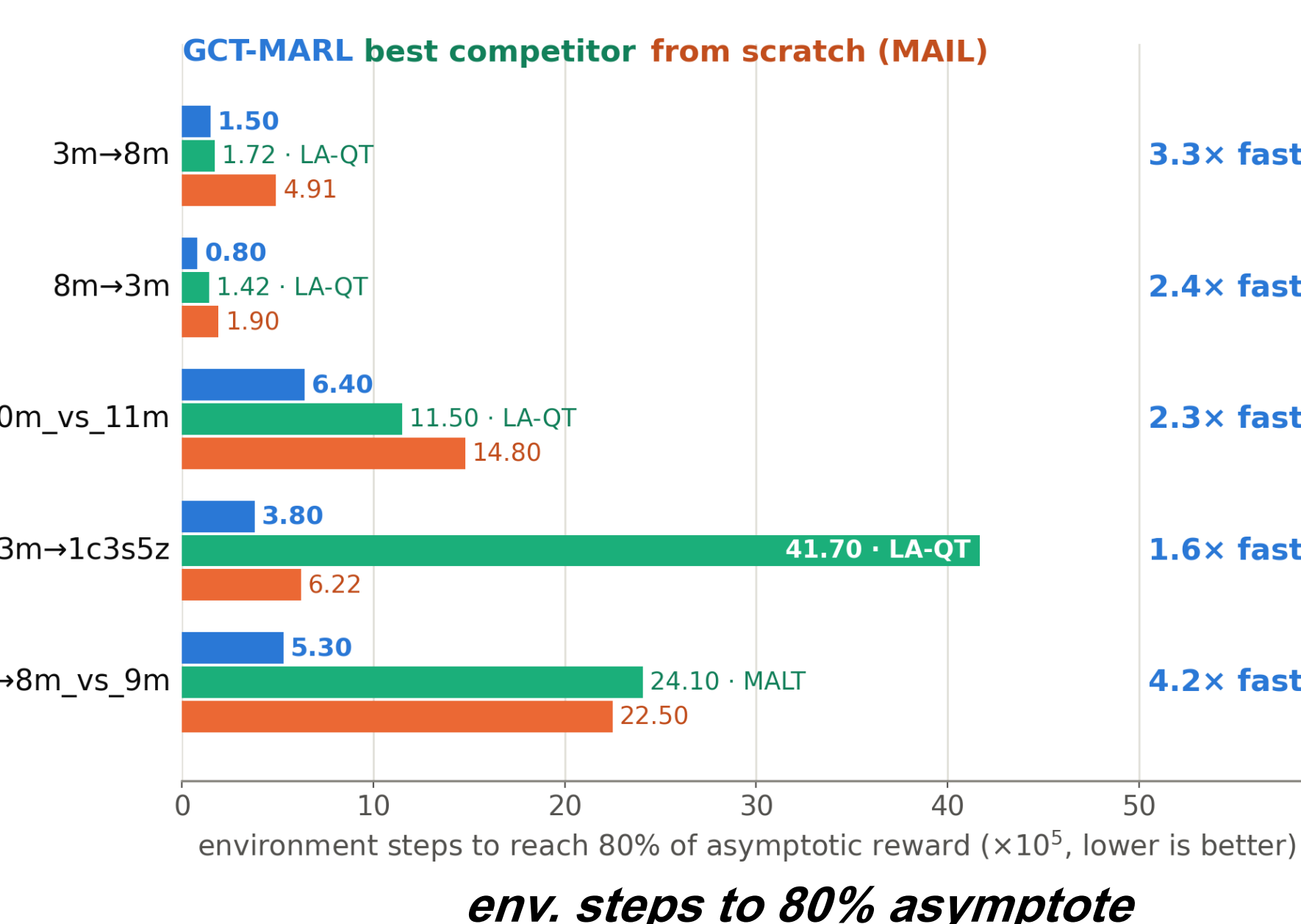
4. Results — SMAC Transfer, View Ablation, Continual Learning



GCT-MARL (blue) vs from scratch (orange) — win rate



learned $\alpha \rightarrow$ topological view



	3m	8m	8m_vs_9m	10m_vs_11m
after phase 1	1.00			
after phase 2	0.91	1.00		
after phase 3	0.72	1.00	0.97	
after phase 4	0.78	0.91	0.91	1.00

diagonal = forward win rate · off-diagonal = retention

continual retention matrix

Takeaways

- ✓ 2–4× faster to the 80% asymptote across five transfer regimes.
- ✓ Learned α collapses onto the topological view; higher-order structure is what transfers.
- ✓ Continual learning: final ACC 0.898, avg BT −0.125.

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For
Code



Continual
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